

A Line-Summand Case of the Convex Supermodularity Question

Candidate partial result for arXiv:1704.05486

Run `fa_banach_001`; generated 2026-07-19

Status

Candidate partial result, likely valid subject to human review. This packet gives a new affirmative subcase of Remark 4.16(1) in [1]. It does not solve the full question where B and C are arbitrary compact subsets of $\mathbb{R}^n$.

Source Metadata

- Matthieu Fradelizi, Mokshay Madiman, Arnaud Marsiglietti, and Artem Zvavitch, “The convexification effect of Minkowski summation,” arXiv:1704.05486, EMS Surveys in Mathematical Sciences 5 (2018), no. 1/2, 1–64 [1].
- Source question location: arXiv PDF page 38, Section 4.2, Remark 4.16(1).
- Local source PDF: `source_paper.pdf`.

Remark 4.16. 1. *One may wonder if Proposition 4.15 extends to higher dimension. More particularly, we do not know if the supermodularity inequality*

$$\text{Vol}_n(A + B + C) + \text{Vol}_n(A) \geq \text{Vol}_n(A + B) + \text{Vol}_n(A + C)$$

holds true in the case where A is convex and B and C are any compact sets.

Figure 1: Remark 4.16(1), asking whether the supermodularity inequality remains true in higher dimension when A is convex and B, C are arbitrary compact sets.

Question Restated

The source paper proves that

$$\text{Vol}_n(A + B + C) + \text{Vol}_n(A) \geq \text{Vol}_n(A + B) + \text{Vol}_n(A + C)$$

for compact convex $A, B, C \subset \mathbb{R}^n$, and also proves the one-dimensional variant

$$\text{Vol}_1(A + B + C) + \text{Vol}_1(\text{conv } A) \geq \text{Vol}_1(A + B) + \text{Vol}_1(A + C)$$

for arbitrary compact $A, B, C \subset \mathbb{R}$. Remark 4.16(1) asks whether, in higher dimension, the first inequality remains true when only A is assumed convex and B, C are arbitrary compact sets.

Proof Intuition

The one-dimensional proof works because, after translating B and C , an intersection point of $A+B$ and $A+C$ is forced to lie between two points of A , hence in $\text{conv } A$. In higher dimension this order argument is lost. However, if B and C both lie on the same line, the missing order reappears on every fiber parallel to that line. A convex set has interval fibers in every direction, so the source paper's one-dimensional inequality can be applied fiber by fiber and then integrated by Fubini.

Main Result

Theorem 1 (fiber-convex line-summand supermodularity). *Let $E \subset \mathbb{R}^n$ be a one-dimensional linear subspace. Let $A \subset \mathbb{R}^n$ be compact and assume that every fiber of A parallel to E is either empty or a compact interval. Let B, C be compact subsets of affine lines parallel to E . Then*

$$\text{Vol}_n(A + B + C) + \text{Vol}_n(A) \geq \text{Vol}_n(A + B) + \text{Vol}_n(A + C).$$

Corollary 1. *In particular, Remark 4.16(1) of [1] has an affirmative answer in all dimensions whenever the arbitrary compact summands B and C are contained in a common affine line. Namely, if $A \subset \mathbb{R}^n$ is compact and convex and B, C are compact subsets of one affine line, then*

$$\text{Vol}_n(A + B + C) + \text{Vol}_n(A) \geq \text{Vol}_n(A + B) + \text{Vol}_n(A + C).$$

Proof

We first record the one-dimensional input, which is Proposition 4.15 of [1] in the special case where the first set is already an interval.

Lemma 1. *Let $I \subset \mathbb{R}$ be a compact interval, possibly empty, and let $S, T \subset \mathbb{R}$ be compact. Then*

$$\text{Vol}_1(I + S + T) + \text{Vol}_1(I) \geq \text{Vol}_1(I + S) + \text{Vol}_1(I + T).$$

Proof. If $I = \emptyset$, the assertion is trivial. Otherwise $\text{conv } I = I$, so this is exactly Proposition 4.15 of [1]. The source proof is elementary: translate S, T so that $\max S = 0 = \min T$, use $(I + S) \cup (I + T) \subset I + S + T$, and observe that $(I + S) \cap (I + T) \subset I$. $\square$

We now prove Theorem 1. By translating B and C independently, we may assume that $B, C \subset E$; all four volumes in the desired inequality are unchanged by such translations. Choose an orthogonal decomposition

$$\mathbb{R}^n = H \oplus E,$$

where $H = E^\perp$, and identify each fiber parallel to E with a subset of $\mathbb{R}$. For $y \in H$, write

$$A_y = \{t \in E : y + t \in A\}.$$

By hypothesis, A_y is empty or a compact interval. Since $B, C \subset E$, the corresponding fibers satisfy

$$(A + B)_y = A_y + B, \quad (A + C)_y = A_y + C, \quad (A + B + C)_y = A_y + B + C.$$

Applying Lemma 1 to the one-dimensional compact interval A_y for every $y \in H$ gives

$$\text{Vol}_1((A + B + C)_y) + \text{Vol}_1(A_y) \geq \text{Vol}_1((A + B)_y) + \text{Vol}_1((A + C)_y).$$

The involved sets are compact, hence Lebesgue measurable, and the section functions are measurable. Integrating this pointwise inequality over H and using Fubini’s theorem yields

$$\text{Vol}_n(A + B + C) + \text{Vol}_n(A) \geq \text{Vol}_n(A + B) + \text{Vol}_n(A + C).$$

This proves the theorem. If A is convex, then every line-parallel fiber of A is an interval, giving the stated corollary. $\square$

Verification Notes

- Local duplicate search: the lightweight indexes were searched for 1704.05486, the title, “convexification effect,” and “Minkowski summation.” No prior exact packet for this paper was found. The only “convexification” hits concerned unrelated arXiv:2305.07169 packets.
- Source check: the target is not answered in the same source beyond the one-dimensional Proposition 4.15 and convex- B, C Theorem 4.5. The present argument is the fiberwise lift of Proposition 4.15 under the extra geometric assumption that B and C are parallel one-dimensional compact sets.
- Exploratory finite-translate searches for square and rectangle A did not find a counterexample to the full question. These searches are not used in the proof; they are documented only as a smoke check in `code/`.

Limitations and Novelty Check

The theorem does not address arbitrary higher-dimensional compact summands B, C . It also does not address the separate fractional-power inequality $\text{Vol}_n^{1/n}$ in Remark 4.16(2), which the source already disproves by an explicit convex-body example.

Bounded novelty checks were performed against the run’s lightweight indexes, the source TeX, and web searches for the exact supermodularity expression. The searches also used phrases such as “volume of Minkowski sums supermodular” and “convexification effect Minkowski summation Remark 4.16.” These checks found the original EMS/arXiv paper and later work on star-shaped sets and fractional Brunn–Minkowski inequalities, but no exact statement of this line-summand fiber theorem. Novelty confidence is therefore bounded rather than exhaustive.

Human Review Recommendation

Review the fiber identity $(A + B + C)_y = A_y + B + C$ after translating B, C into the common line, and check that Fubini applies to the compact Minkowski sums. If those standard measurability points pass, the partial result should be accepted as a clean affirmative slice of Remark 4.16(1).

References

- [1] M. Fradelizi, M. Madiman, A. Marsiglietti, and A. Zvavitch, *The convexification effect of Minkowski summation*, EMS Surveys in Mathematical Sciences 5 (2018), no. 1/2, 1–64. arXiv:1704.05486. DOI: 10.4171/EMSS/26.